

STEM Reasoning Lab

Mixed Assessment - Questions and rubric signals

Q1

Calculus

5 points

Evaluate $\int_0^1 x \exp(x^2) dx$. Show the substitution and the transformed bounds.

RUBRIC SIGNALS

- $u = x^2$
- the $1/2$ factor
- bounds and final evaluation

Q2

Mechanics

8 points

A 2 kg block slides 3 m from rest down a 30 degree incline with $\mu_k = 0.20$. Find its acceleration and final speed. Use $g = 9.81 \text{ m/s}^2$.

RUBRIC SIGNALS

- force resolution
- friction direction
- acceleration
- speed and units

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Q3

Linear algebra proof

7 points

For any real $m \times n$ matrix A , prove $\ker(A) = \ker(A^T A)$, then conclude $\text{rank}(A) = \text{rank}(A^T A)$. Do not assume A is square or invertible.

RUBRIC SIGNALS

- forward inclusion
- norm identity for the reverse inclusion
- rank-nullity

REFERENCE IDENTITY

$$x^T A^T A x = (Ax)^T (Ax) = \|Ax\|^2$$

The identity is supplied as a grading reference, not as part of the student prompt.

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Q4

Programming

10 points

Implement `stable_softmax(xs)`. Return `[]` for empty input and avoid overflow for values near 1000. Do not use NumPy.

RUBRIC SIGNALS

- empty input
- subtract maximum
- normalization
- extreme-value behavior

```
[]                -> []  
[0, 0]            -> [0.5, 0.5]  
[1, 2, 3]         -> [0.09003, 0.24473, 0.66524]  
[1000, 1000]      -> [0.5, 0.5]  
[-1000, 0, 1000] -> approximately [0, 0, 1]
```

The public demo shows frozen test outcomes; it never executes visitor code.